



Virtual Glasses Try-On & Sizing Assistant

Semester Project

by

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Subject : Entrepreneurship

Introduction



- The eyewear market is evolving with growing demand for virtual try-on solutions.
- Accurate frame sizing and shape matching remain a challenge in online and in-store shopping.
- Incorrect frame fit leads to discomfort, increased returns, and customer dissatisfaction.
- This project develops a real-time virtual glasses try-on and sizing assistant t
- Measures key facial landmarks with precision
- Classifies face shape to guide frame style selection
- Recommends optimal frame sizes for a comfortable fit
- Enables interactive, real-time frame try-on with live webcam feed

Tools



- Python Core programming language for application logic
- PyQt5 GUI framework for building the interactive desktop interface
- OpenCV Real-time video capture and image processing
- Media Pipe Face Mesh Accurate facial landmark detection
- NumPy & SciPy Numerical computations and geometric calculations
- Multithreading (QThread) Smooth video streaming and processing without UI freezes

Tools



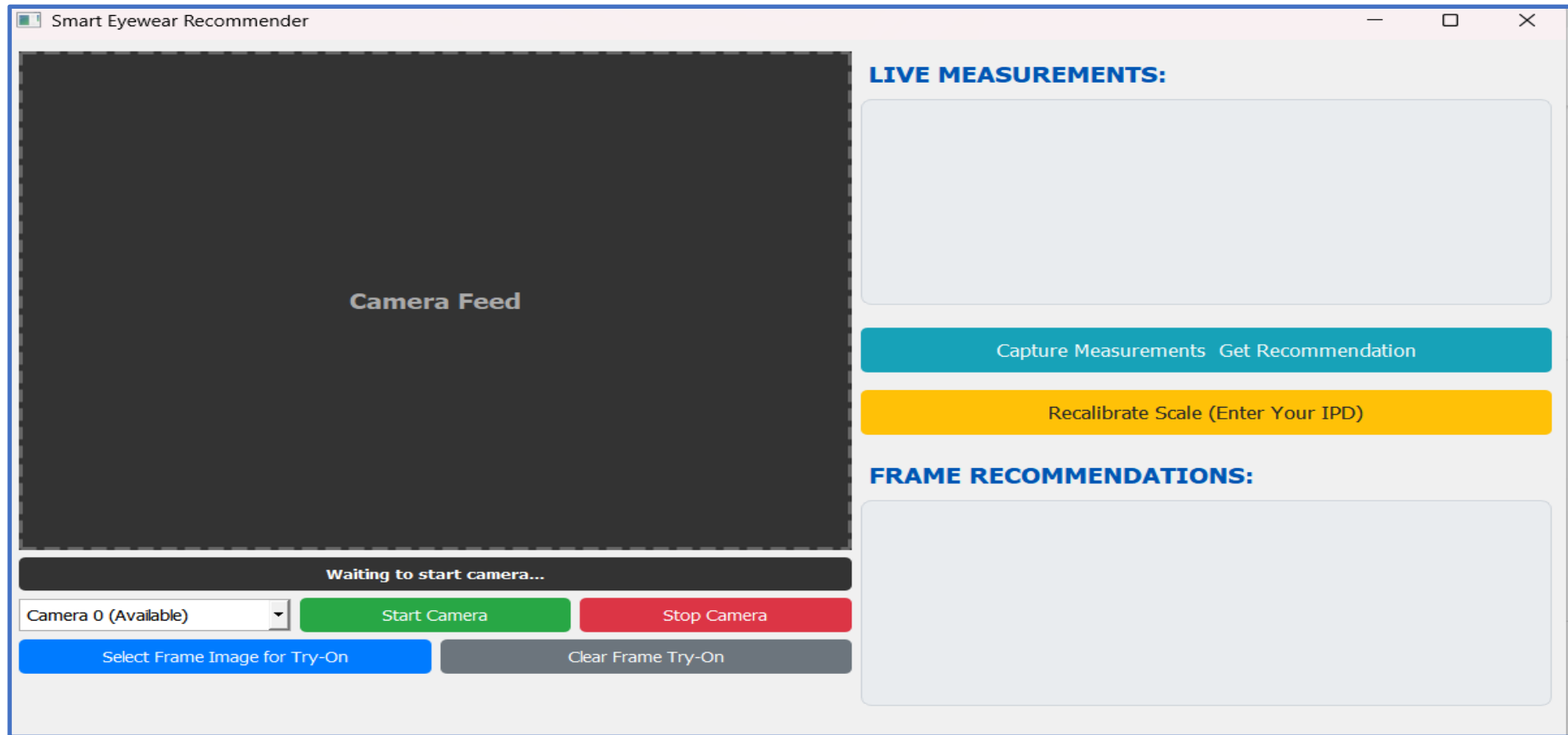
- QImage & QPixmap (Qt) Displaying processed frames and overlays
- Custom Algorithms

Face shape classification based on geometric features

Frame size and style recommendation system

- Dialogs & User Interaction Custom dialogs for user inputs (e.g., IPD calibration)
- File Dialogs Frame image selection and management
- Used AI for the implementation of this logic (ChatGPT, Grok, Gemini and Blackbox)

GUI (Features)



Target Market



Our application targets:

- Online Eyewear Retailers To integrate with their websites and reduce return rates.
- Direct Consumers Individuals who buy glasses online and want to ensure a perfect fit.
- Optical Stores & Brands To enhance their online shopping experience with AI-based recommendations.

Business model



1. B2B (Business-to-Business):

- Partnering with eyewear brands and online retailers.
- Offering API integration for virtual try-ons on e-commerce platforms.
- Subscription-based model for businesses.

2. B2C (Business-to-Consumer):

- Free mobile/web app with limited features.
- Premium subscription for advanced features (e.g., expert recommendations, more frame options).

Demonstration

